TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
System of inequalities: graph and shade each one; the overlap is the solution set.
A solution makes EVERY inequality true. Points on a dashed line are not solutions.
In a story: no negative amounts, so stay in the first quadrant.

1. Is (0, 0) a solution of $x + y > 1$?

Tip: substitute (0, 0)

A. No, because $0 > 1$ is false

B. Yes, because $0 < 1$

C. Yes, because $0 = 0$

2. Where is the solution set of $y > x + 2$ and $y > x + 5$?

Tip: both $>$ → above both

F. below $y = x + 2$

G. above $y = x + 5$

H. between the lines

3. The lines of $y > -x + 1$ and $y \leq x + 3$ cross at $(-1, 2)$. Is $(-1, 2)$ a solution?

> **Tip:** check every inequality

A. Yes, because it is where the lines meet

B. Yes, because $2 \leq 2$

C. No, because $2 > 2$ is false

4. $x + y \leq 45$ (minutes). Which pair works?

> **Tip:** add, then compare

F. (30, 20)

G. (45, 1)

H. (20, 25)